

Relevance-Based Prediction: Factor Selection for Saugus Schools

Cross-section year: FY2024 — the latest year with complete MCAS outcomes and DESE per-pupil spending data

Four outcomes: MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, Education Budget Share

One RBP run per outcome over the full candidate pool — no in-model pruning (faithful to Kritzman)

The one discretionary step is which factors enter and how they tier: Tier 3 (structural) matches Saugus to peers; Tiers 1 & 2 (actionable) are Saugus-specific

Saugus analyzed as the prediction task; importance flags where it is most distinctive among actionable factors — a prediction-share metric

Leave-one-out LOO r validates the predictions across all MA districts

Model Summary (all candidates used — no pruning)			
MCAS Grades 3–8	MCAS Grade 10 (ELA)	Education Budget Share	Dropout Rate
Predicted on 19 factors	Predicted on 19 factors	Predicted on 19 factors	Predicted on 19 factors
LOO r = 0.90 · R² = 0.75	LOO r = 0.90 · R² = 0.76	LOO r = 0.80 · R² = 0.64	LOO r = 0.84 · R² = 0.64
Typical error ± 5.6 pp	Typical error ± 6.6 pp	Typical error ± 3.9 pp	Typical error ± 0.7 pp

cted vs. actual (range −1 to +1; noise below ≈ 0.15 at this sample size). R² = 1 − SSE/SS1, the out-of-sample share of town-to-town variance explained (≤ r² because RBP’s weighted averages compress toward the mean). Typical e

How to Read This Report — Three Tiers of Factors

This report matches Saugus to demographically-similar towns using Tier 3, then ranks the Tier 1 & 2 factors Saugus can actually change. Structural traits are the backdrop for finding peers — never a recommendation.

TIER 1 — Directly votable (Town Meeting / ballot)

What the town changes by a vote.

- Education's share of the municipal budget
- Residential tax rate · Proposition 2½ override · debt exclusion
- Spending above the state-required school minimum (Ch. 70)
- Reserves (free cash, stabilization) · OPEB funding · capital projects

TIER 2 — Policy / management (administration decides)

What the school department & town administration run day-to-day (funded by Tier 1).

- Chronic absenteeism — attendance & re-engagement programs
- Teacher staffing levels and pay
- How the school dollar splits between classroom and overhead
- Above-minimum school spending funded from reserves

TIER 3 — Structural (NOT changeable by vote)

What the community IS — used ONLY to find Saugus's true peer towns.

- Household income · property wealth · adult education
- Poverty · English-learner · special-education shares
- Housing tenure · district enrollment · regional economy
- → Defines the peer group; never shown as an actionable factor here.

How RBP Compares Saugus to Towns Like It — Without Shrinking the Data

It focuses on towns similar to Saugus using the FULL set of MA districts, never a hand-picked subset — so the comparison keeps its statistical power.

1. Full sample, always — no pre-filtering

Every prediction uses all ~170–220 MA districts.

The full sample is what learns which towns are relevant to Saugus (the factor covariance Ω). Hand-filtering to a small peer set first would make that geometry unestimable and turn the importance scores into noise.

2. Signed weighting — contrasts, not exclusion

Each town's weight may be positive OR negative, and the weights sum to 1.

Most-relevant towns get large positive weights; less-relevant towns get small, often negative weights — used as contrasts (subtracted), as in regression. Only ~6–15 of ~170–220 towns sit near zero, and none is excluded.

3. A small effective core, plus a censored cross-check

About half the towns carry small negative weights; ~10–25 carry most of the weight.

RBP also re-predicts from only the top 80% / 50% / 20% most-relevant towns and blends those tight-peer estimates by how reliably each predicts (its fit) — the method decides how much to trust the tight group; no human picks the cutoff.

The math behind it

$$\hat{y}_{Saugus} = \sum_i w_i y_i, \quad \sum_i w_i = 1$$

$$w_i = \frac{1}{N} + \frac{\lambda^2}{n-1} (\delta_i r_i - \phi \bar{r})$$

$$r_i = -\frac{1}{2} (x_i - x_S)^T \Omega^{-1} (x_i - x_S) + \text{info}$$

$$N_{\text{eff}} = 1 / \sum_i w_i^2$$

$$w = x_S^T (X^T X)^{-1} X^T$$

The prediction is a signed weighted sum of town outcomes; the weights sum to 1.

Each weight = a uniform 1/N baseline + a tilt that turns NEGATIVE below average relevance.

Relevance r = closeness to Saugus in factor space (Mahalanobis); Ω estimated from all N towns.

Effective # of towns carrying the weight ≈ 10 –25 of ~170–220 — a focus, not a deletion.

With all factors and no censoring this is exactly OLS — signed weights are inherent to regression.

Because the relevance geometry Ω is learned from the full sample and the weights sum to 1, concentrating the prediction on a relevant core costs no stability — the instability a small hand-picked subset would create.

Combined Model Summary: All Four Outcomes

RBP factor selection results across MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, and Education Budget Share

Model	Predicted	Actual	Gap	Direction	LOO r	Factors
MCAS Grades 3–8	23.6%	29.0%	+5.4pp	over	+0.902	19
MCAS Grade 10 (ELA)	34.9%	47.0%	+12.1pp	over	+0.900	19
Education Budget Share	40.9%	32.3%	-8.6pp	under	+0.801	19
Dropout Rate	4.1%	2.9%	-1.2pp	over	+0.838	19

Actionable Factor Cross-Reference — Which Factors Matter Across Outcomes

Actionable factors only (structural traits are matching-only). A factor that matters in multiple outcomes is a robust target.

Factor	# Models	MCAS 3–8	MCAS 10	Ed Budget	Dropout
Students chronically absent (10%+)	4	+0.541	+0.622	-0.026	+0.908
School spend vs. property wealth	4	+0.306	+0.442	+0.270	+0.282
Teacher share of school spending	4	+0.378	+0.361	+0.175	+0.327
Teachers per low-income student	4	+0.375	+0.352	+0.127	+0.335
Teachers per 100 students	4	+0.359	+0.346	+0.079	+0.387
Residential tax rate (per \$1,000)	4	+0.322	+0.349	+0.149	+0.325
Spending vs Ch70 required minimum	4	+0.377	+0.374	+0.064	+0.314
Net school spending per pupil	4	+0.326	+0.357	+0.047	+0.233
Ed budget share	3	-0.116	-0.163	—	-0.218
Fixed costs (mostly health ins.)	1	—	—	+0.393	—

Gap = actual – predicted; the Direction column resolves the sign (for dropout, lower-than-predicted is an over-performance). Green rows = factor matters in 3+ outcomes. Cell colour: green = positive importance, red = NEGATIVE importance — the factor adds noise rather than signal for that outcome (Kritzman fn. 12; expected, not an error) — white = not a candidate in that model (—).

What This Means: MCAS Grades 3-8

Predicted 23.6pp · Actual 29.0pp · Gap +5.4pp (higher is better) · Saugus ranks 54 of 219 MA districts on this measure

The Bottom Line

Saugus's MCAS grades 3–8 proficiency (ELA + Math, meeting/exceeding) is 29.0%, 5.4pp better than its demographic prediction of 23.6% — ranking 54 of 219 MA districts, in the top 25%. A district with Saugus's income, poverty, and enrollment profile is predicted to land near this level.

Actionable Factors Where Saugus Stands Out

Ranked by RBP importance — where Saugus is most distinctive, Saugus vs. the statewide median

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.54
Teacher share of school spending	0.349	0.368	+0.38
Spending vs Ch70 required minimum	1.03	1.36	+0.38
Teachers per low-income student	0.126	0.277	+0.37
Teachers per 100 students	5.87	8.07	+0.36

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad, and not how much changing a factor would move the outcome. For where to act, read the peer comparison at right: the factors on which Saugus differs most from comparable towns that did better.

What Comparable Better Towns Do Differently

Same factors, for the 14 towns the model predicted like Saugus that scored better:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	20.5%	-10.7pp
Teacher share of school spending	0.349	0.366	+0.016
Spending vs Ch70 required minimum	1.03	1.16	+0.130
Teachers per low-income student	0.126	0.167	+0.041
Teachers per 100 students	5.87	7.57	+1.70

Difference = comparable-town median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

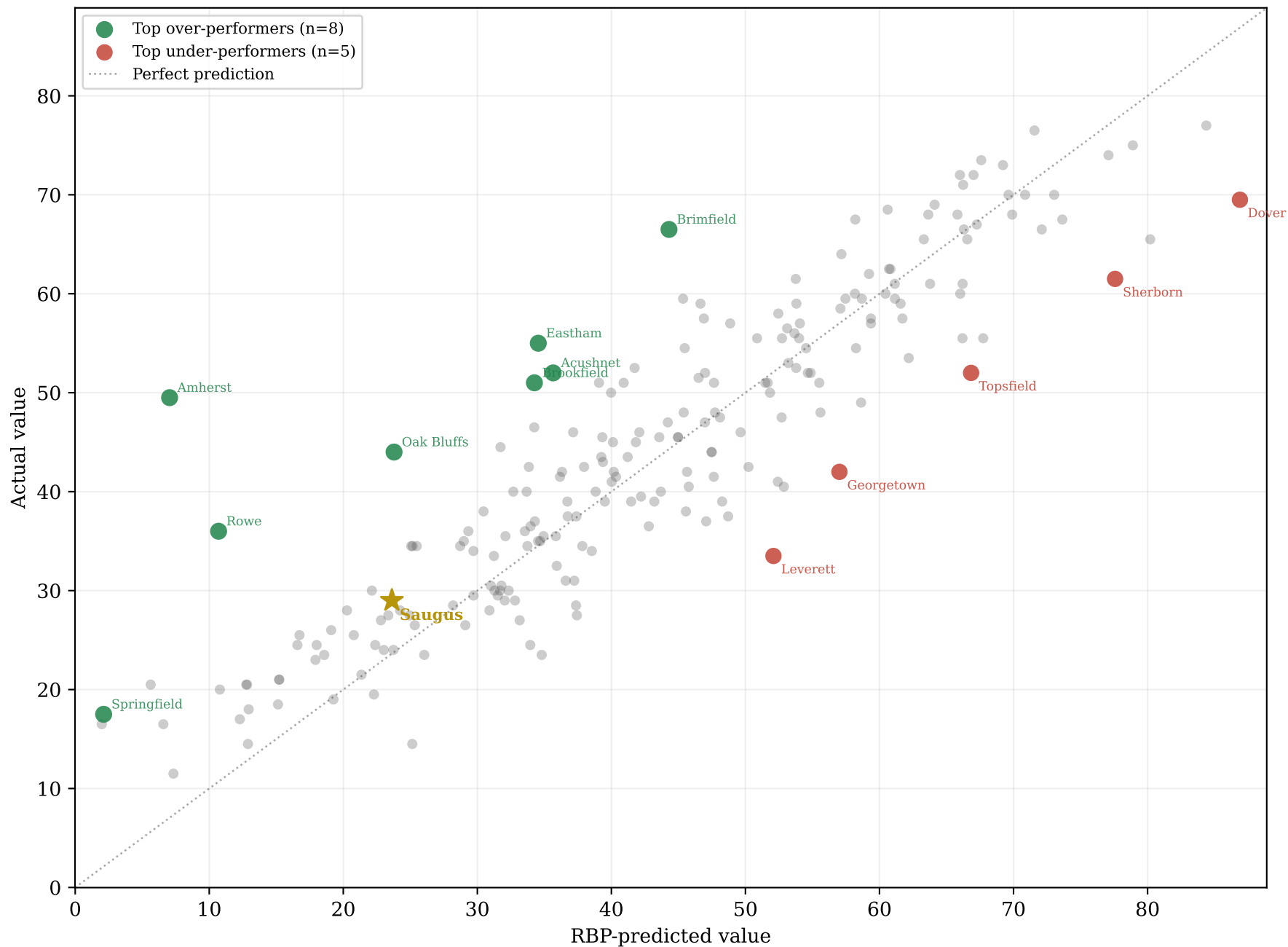
Saugus does better here than its demographic profile predicts. The factor on which Saugus is most demographically unusual is students chronically absent (10%+) (31.2% vs the 15.8% state median; peer gap -10.7pp). It both tops the RBP importance and shows the widest peer gap — two views of the same outlier status, not independent confirmation. (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) The factors below are where Saugus differs most from the towns that beat their own predictions — the most plausible places to focus attention, though the comparison is associational, not proof of cause.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns the model predicted like Saugus that scored better. Structural traits are used for peer-matching only.

Over- and Under-Performers: MCAS Grades 3-8

Saugus gap: +5.4pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grades 3-8 (LOO r = 0.902)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Comparable Better Towns Do Differently: MCAS Grades 3-8

Towns the model predicted like Saugus that scored better — their actionable-factor values vs Saugus, ordered by where Saugus is furthest from these peers (gap measured in standard deviations, so different-unit factors compare)

Saugus (highlighted) vs towns predicted like Saugus that scored better
Ordered by largest peer gap (in SDs), top 9 shown

Factor	Saugus	Oak Bluffs	West Springfield	Peer med	Imp.
Students chronically absent (10%+)	31.2	17.8	20.9	20.5	+0.541
Ed budget share	32.3	40.0	40.9	41.4	-0.116
Residential tax rate (per \$1,000)	10.7	5.23	14.8	13.8	+0.322
Teachers per 100 students	5.87	11.5	8.09	7.57	+0.359
School spend vs. property wealth	0.07	0.03	0.14	0.12	+0.306
Net school spending per pupil	17,369	30,814	18,761	20,632	+0.326
Spending vs Ch70 required minimum	1.03	1.71	1.06	1.16	+0.377
Teacher share of school spending	0.35	0.41	0.38	0.37	+0.378
Teachers per low-income student	0.13	0.31	0.14	0.17	+0.375

Predicted like Saugus — but scored better

District	Actual	Predicted	vs Expected
>> Saugus	29.0	23.6	+5.4pp
Oak Bluffs	44.0	23.8	+20.2pp
West Springfield	34.5	25.1	+9.4pp
Clinton	30.0	22.1	+7.9pp
Leominster	34.5	25.2	+9.3pp
Malden	34.5	25.5	+9.0pp
Westfield	34.5	28.7	+5.8pp
Easthampton	35.0	29.0	+6.0pp
Holbrook	36.0	29.3	+6.7pp
Edgartown	34.0	29.7	+4.3pp
Monson	29.5	29.7	-0.2pp
Rockland	38.0	30.5	+7.5pp
Gloucester	30.5	31.0	-0.5pp

(closest 12 shown; + 2 more comparable towns within the model's prediction band)

Peer med = median across ALL comparable better-performing towns (not just the examples shown). vs Expected = how far actual beats the demographic prediction, signed so + always means better than expected (for dropout, lower-than-predicted shows as a positive 'better'). Factor values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grade 10 (ELA)

Predicted 34.9pp · Actual 47.0pp · Gap +12.1pp (higher is better) · Saugus ranks 10 of 168 MA districts on this measure

The Bottom Line

Saugus's grade 10 ELA proficiency is 47.0%, 12.1pp better than its demographic prediction of 34.9% — ranking 10 of 168 MA districts, in the top 10% of MA districts. Note: the grade 3–8 academic base is an outcome, not an actionable factor, so it is excluded from this model — part of the over-performance shown here reflects that omitted prior strength rather than a grade-10 effect.

Actionable Factors Where Saugus Stands Out

Ranked by RBP importance — where Saugus is most distinctive, Saugus vs. the statewide median

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.62
School spend vs. property wealth	0.068	0.082	+0.44
Spending vs Ch70 required minimum	1.03	1.36	+0.37
Teacher share of school spending	0.349	0.368	+0.36
Net school spending per pupil	\$17,369	\$20,865	+0.36

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad, and not how much changing a factor would move the outcome. For where to act, read the peer comparison at right: the factors on which Saugus differs most from comparable towns that did better.

What Comparable Better Towns Do Differently

Same factors, for the 4 towns the model predicted like Saugus that scored better:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	23.1%	-8.1pp
School spend vs. property wealth	0.068	0.112	+0.045
Spending vs Ch70 required minimum	1.03	1.09	+0.054
Teacher share of school spending	0.349	0.375	+0.026
Net school spending per pupil	\$17,369	\$20,335	+2,966

Difference = comparable-town median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

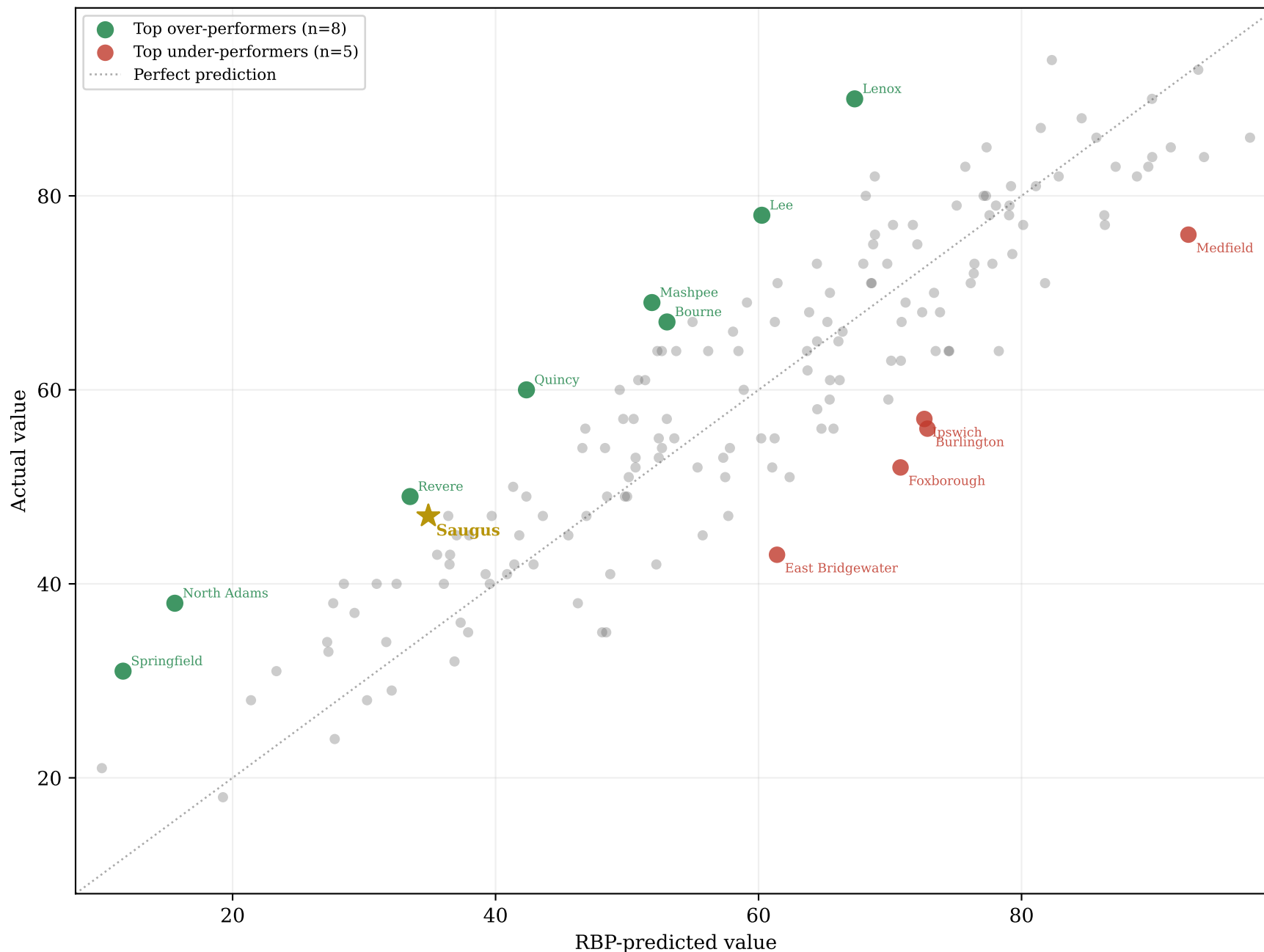
Saugus does better here than its demographic profile predicts. The strongest actionable factor is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the comparable towns that scored better is net school spending per pupil (+2,966, peer median – Saugus). (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) Among the actionable factors, chronic absenteeism and teacher staffing are the most visible — Saugus clears this bar despite lagging peers on both.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns the model predicted like Saugus that scored better. Structural traits are used for peer-matching only.

Over- and Under-Performers: MCAS Grade 10 (ELA)

Saugus gap: +12.1pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grade 10 (ELA) (LOO r = 0.900)



Residual = actual – predicted. Positive = performing better than demographics suggest.

What Comparable Better Towns Do Differently: MCAS Grade 10 (ELA)

Towns the model predicted like Saugus that scored better — their actionable-factor values vs Saugus, ordered by where Saugus is furthest from these peers (gap measured in standard deviations, so different-unit factors compare)

Saugus (highlighted) vs towns predicted like Saugus that scored better
Ordered by largest peer gap (in SDs), top 9 shown

Factor	Saugus	Revere	Salem	Peer med	Imp.
Ed budget share	32.3	43.4	43.4	43.3	-0.163
Teachers per 100 students	5.87	7.89	7.97	7.70	+0.346
Students chronically absent (10%+)	31.2	30.0	20.3	23.1	+0.622
School spend vs. property wealth	0.07	0.10	0.13	0.11	+0.442
Net school spending per pupil	17,369	20,324	25,821	20,335	+0.357
Teacher share of school spending	0.35	0.40	0.34	0.38	+0.361
Residential tax rate (per \$1,000)	10.7	9.11	11.6	11.4	+0.349
Spending vs Ch70 required minimum	1.03	1.02	1.39	1.09	+0.374
Teachers per low-income student	0.13	0.12	0.13	0.12	+0.352

Predicted like Saugus — but scored better

District	Actual	Predicted	vs Expected
>> Saugus	47.0	34.9	+12.1pp
Revere	49.0	33.5	+15.5pp
Salem	50.0	41.3	+8.7pp
Malden	49.0	42.3	+6.7pp
Quincy	60.0	42.3	+17.7pp

Peer med = median across ALL comparable better-performing towns (not just the examples shown). vs Expected = how far actual beats the demographic prediction, signed so + always means better than expected (for dropout, lower-than-predicted shows as a positive 'better'). Factor values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Education Budget Share

Predicted 40.9pp · Actual 32.3pp · Gap -8.6pp (higher is better) · Saugus ranks 214 of 219 MA districts on this measure

The Bottom Line

Saugus's education budget share is 32.3%, 8.6pp worse than its demographic prediction of 40.9% — ranking 214 of 219 MA districts, in the bottom 10%. Unlike the academic measures, this gap reflects a municipal budgeting choice, not a demographic constraint. Note that the budget shares are compositional — they sum to the total budget — so a high importance for a competing category is partly mechanical (a larger slice elsewhere necessarily leaves a smaller one for schools), not an independent effect.

Actionable Factors Where Saugus Stands Out

Ranked by RBP importance — where Saugus is most distinctive, Saugus vs. the statewide median

Factor	Saugus	MA median	Importance
Fixed costs (mostly health ins.)	25.3%	17.5%	+0.39
School spend vs. property wealth	0.068	0.082	+0.27
Teacher share of school spending	0.349	0.368	+0.18
Residential tax rate (per \$1,000)	10.7	12.6	+0.15
Teachers per low-income student	0.126	0.277	+0.13

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad, and not how much changing a factor would move the outcome. For where to act, read the peer comparison at right: the factors on which Saugus differs most from comparable towns that did better.

What Comparable Better Towns Do Differently

Same factors, for the 76 towns the model predicted like Saugus that scored better:

Factor	Saugus	Peer median	Difference
Fixed costs (mostly health ins.)	25.3%	20.4%	-4.9pp
School spend vs. property wealth	0.068	0.091	+0.023
Teacher share of school spending	0.349	0.365	+0.015
Residential tax rate (per \$1,000)	10.7	11.4	+0.760
Teachers per low-income student	0.126	0.203	+0.077

Difference = comparable-town median – Saugus. Sign direction depends on the factor; see takeaway below.

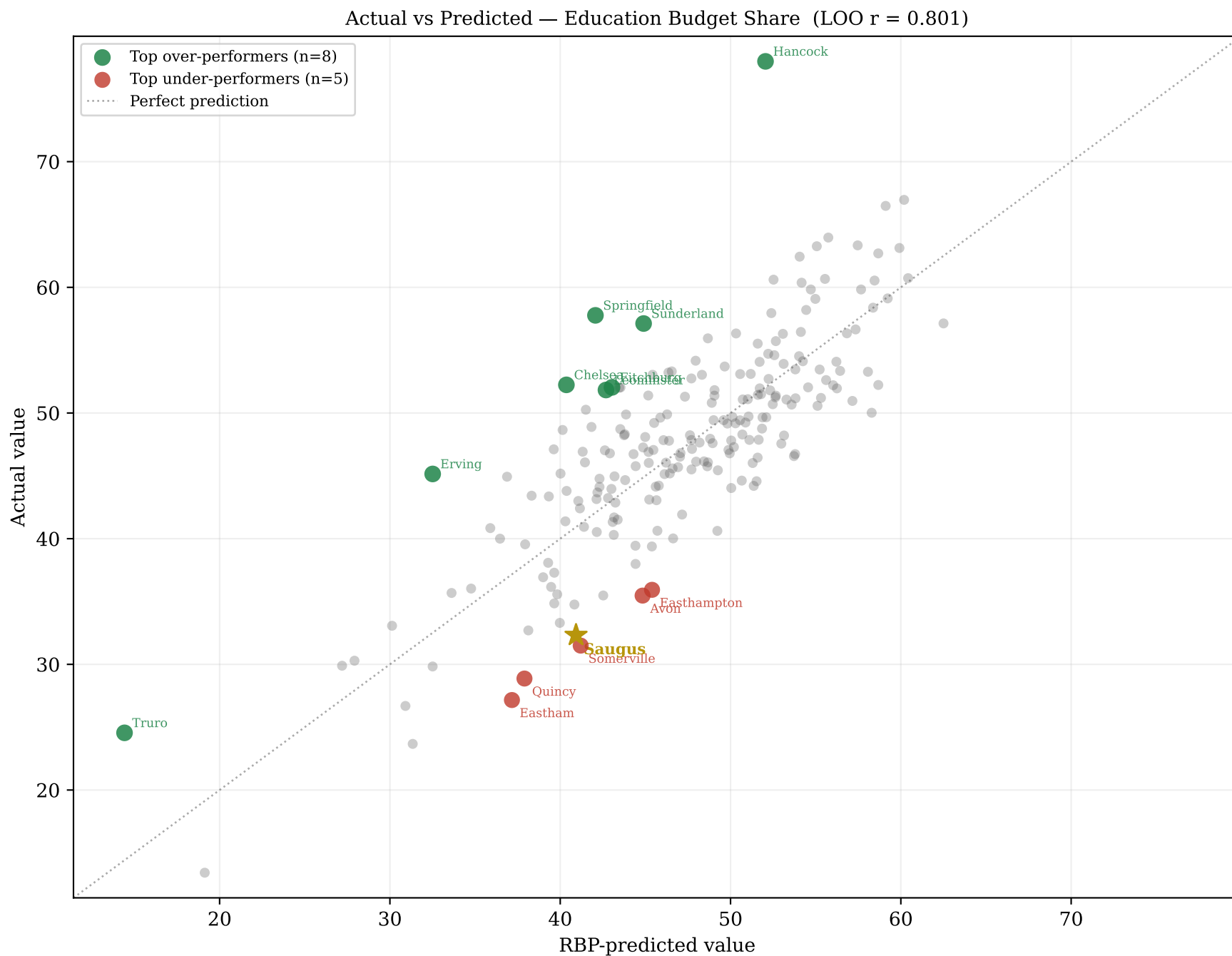
The Takeaway

Saugus does worse here than its demographic profile predicts. The factor on which Saugus is most demographically unusual is fixed costs (mostly health ins.) (25.3% vs the 17.5% state median; peer gap -4.9pp). It both tops the RBP importance and shows the widest peer gap — two views of the same outlier status, not independent confirmation. (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) The top factors are competing budget categories that crowd out the education share — the choice here sits with Town Meeting and municipal budgeting, not the school district.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns the model predicted like Saugus that scored better. Structural traits are used for peer-matching only.

Over- and Under-Performers: Education Budget Share

Saugus gap: -8.6pp · Gold star = Saugus · Green = over-performers · Red = under-performers



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Comparable Better Towns Do Differently: Education Budget Share

Towns the model predicted like Saugus that scored better — their actionable-factor values vs Saugus, ordered by where Saugus is furthest from these peers (gap measured in standard deviations, so different-unit factors compare)

Saugus (highlighted) vs towns predicted like Saugus that scored better
Ordered by largest peer gap (in SDs), top 9 shown

Factor	Saugus	Greenfield	Marlborough	Peer med	Imp.
Students chronically absent (10%+)	31.2	27.8	23.5	18.1	-0.026
Teachers per 100 students	5.87	6.94	7.36	7.92	+0.079
Fixed costs (mostly health ins.)	25.3	23.2	22.3	20.4	+0.393
Net school spending per pupil	17,369	24,237	20,169	22,267	+0.047
Spending vs Ch70 required minimum	1.03	1.42	1.11	1.31	+0.064
School spend vs. property wealth	0.07	0.19	0.09	0.09	+0.270
Teacher share of school spending	0.35	0.28	0.37	0.36	+0.175
Residential tax rate (per \$1,000)	10.7	20.4	10.2	11.4	+0.149
Teachers per low-income student	0.13	0.11	0.13	0.20	+0.127

Predicted like Saugus — but scored better

District	Actual	Predicted	vs Expected
>> Saugus	32.3	40.9	-8.6pp
Greenfield	34.8	40.8	-6.1pp
Marlborough	43.0	41.1	+1.9pp
Plymouth	42.4	41.2	+1.3pp
Beverly	46.9	41.3	+5.6pp
West Springfield	40.9	41.4	-0.5pp
Brewster	46.1	41.5	+4.6pp
Stoughton	43.8	40.4	+3.4pp
Chelsea	52.2	40.4	+11.9pp
Webster	50.3	41.5	+8.8pp
Everett	41.4	40.3	+1.1pp
Lenox	48.6	40.1	+8.5pp
Lee	48.9	41.8	+7.1pp

(closest 12 shown; + 64 more comparable towns within the model's prediction band)

Peer med = median across ALL comparable better-performing towns (not just the examples shown). vs Expected = how far actual beats the demographic prediction, signed so + always means better than expected (for dropout, lower-than-predicted shows as a positive 'better'). Factor values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Dropout Rate

Predicted 4.1pp · Actual 2.9pp · Gap -1.2pp (lower is better) · Saugus ranks 27 of 168 MA districts on this measure

The Bottom Line

Saugus's dropout rate is 2.9%, 1.2pp better than its demographic prediction of 4.1% — ranking 27 of 168 MA districts, in the top 25%. Dropout reflects re-engagement and attendance recovery as much as demographics.

Actionable Factors Where Saugus Stands Out

Ranked by RBP importance — where Saugus is most distinctive, Saugus vs. the statewide median

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.91
Teachers per 100 students	5.87	8.07	+0.39
Teachers per low-income student	0.126	0.277	+0.33
Teacher share of school spending	0.349	0.368	+0.33
Residential tax rate (per \$1,000)	10.7	12.6	+0.33

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad, and not how much changing a factor would move the outcome. For where to act, read the peer comparison at right: the factors on which Saugus differs most from comparable towns that did better.

What Comparable Better Towns Do Differently

Same factors, for the 10 towns the model predicted like Saugus that scored better:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	25.4%	-5.8pp
Teachers per 100 students	5.87	7.46	+1.59
Teachers per low-income student	0.126	0.125	-0.001
Teacher share of school spending	0.349	0.360	+0.011
Residential tax rate (per \$1,000)	10.7	14.1	+3.49

Difference = comparable-town median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

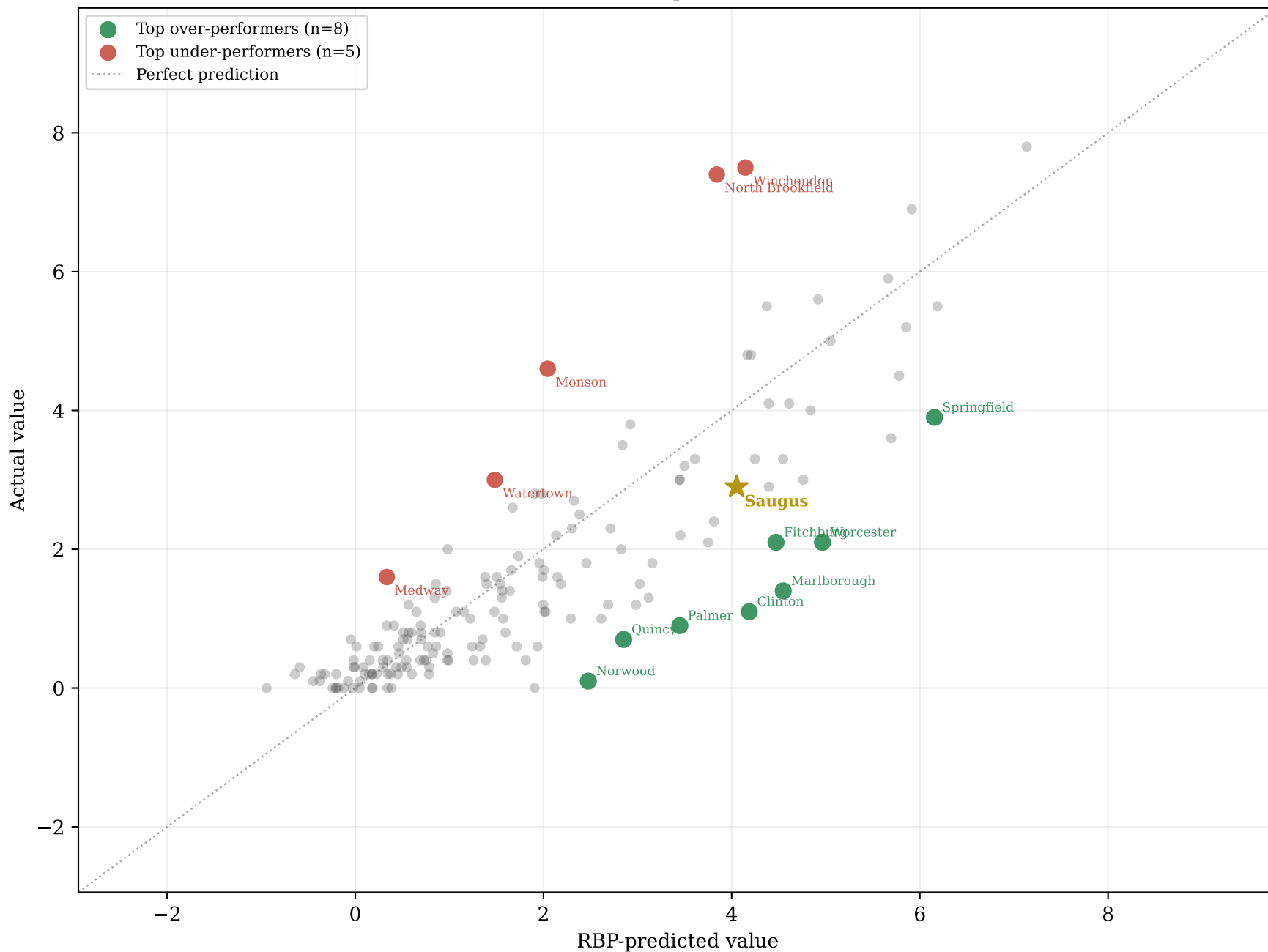
Saugus does better here than its demographic profile predicts. The factor on which Saugus is most demographically unusual is students chronically absent (10%+) (31.2% vs the 15.8% state median; peer gap -5.8pp). It both tops the RBP importance and shows the widest peer gap — two views of the same outlier status, not independent confirmation. (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) Chronic absenteeism is the factor to watch — the main risk to this result if it climbs.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns the model predicted like Saugus that scored better. Structural traits are used for peer-matching only.

Over- and Under-Performers: Dropout Rate

Saugus gap: -1.2pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — Dropout Rate (LOO r = 0.838)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Comparable Better Towns Do Differently: Dropout Rate

Towns the model predicted like Saugus that scored better — their actionable-factor values vs Saugus, ordered by where Saugus is furthest from these peers (gap measured in standard deviations, so different-unit factors compare)

Saugus (highlighted) vs towns predicted like Saugus that scored better
Ordered by largest peer gap (in SDs), top 9 shown

Factor	Saugus	Clinton	Greenfield	Peer med	Imp.
Ed budget share	32.3	49.6	34.8	46.9	-0.218
School spend vs. property wealth	0.07	0.12	0.19	0.16	+0.282
Residential tax rate (per \$1,000)	10.7	13.1	20.4	14.1	+0.325
Teachers per 100 students	5.87	7.31	6.94	7.46	+0.387
Net school spending per pupil	17,369	18,512	24,237	21,582	+0.233
Students chronically absent (10%+)	31.2	7.70	27.8	25.4	+0.908
Spending vs Ch70 required minimum	1.03	1.08	1.42	1.16	+0.314
Teacher share of school spending	0.35	0.39	0.28	0.36	+0.327
Teachers per low-income student	0.13	0.14	0.11	0.12	+0.335

Predicted like Saugus — but scored better

District	Actual	Predicted	vs Expected
>> Saugus	2.9	4.1	+1.2pp better
Clinton	1.1	4.2	+3.1pp better
Greenfield	2.4	3.8	+1.4pp better
Chicopee	2.1	3.8	+1.7pp better
Fitchburg	2.1	4.5	+2.4pp better
Marlborough	1.4	4.5	+3.1pp better
Taunton	2.2	3.5	+1.3pp better
Palmer	0.9	3.4	+2.5pp better
Salem	1.8	3.2	+1.4pp better
Worcester	2.1	5.0	+2.9pp better
Agawam	1.3	3.1	+1.8pp better

Peer med = median across ALL comparable better-performing towns (not just the examples shown). vs Expected = how far actual beats the demographic prediction, signed so + always means better than expected (for dropout, lower-than-predicted shows as a positive 'better'). Factor values shown in raw units; importance scores from RBP Exhibit 5.

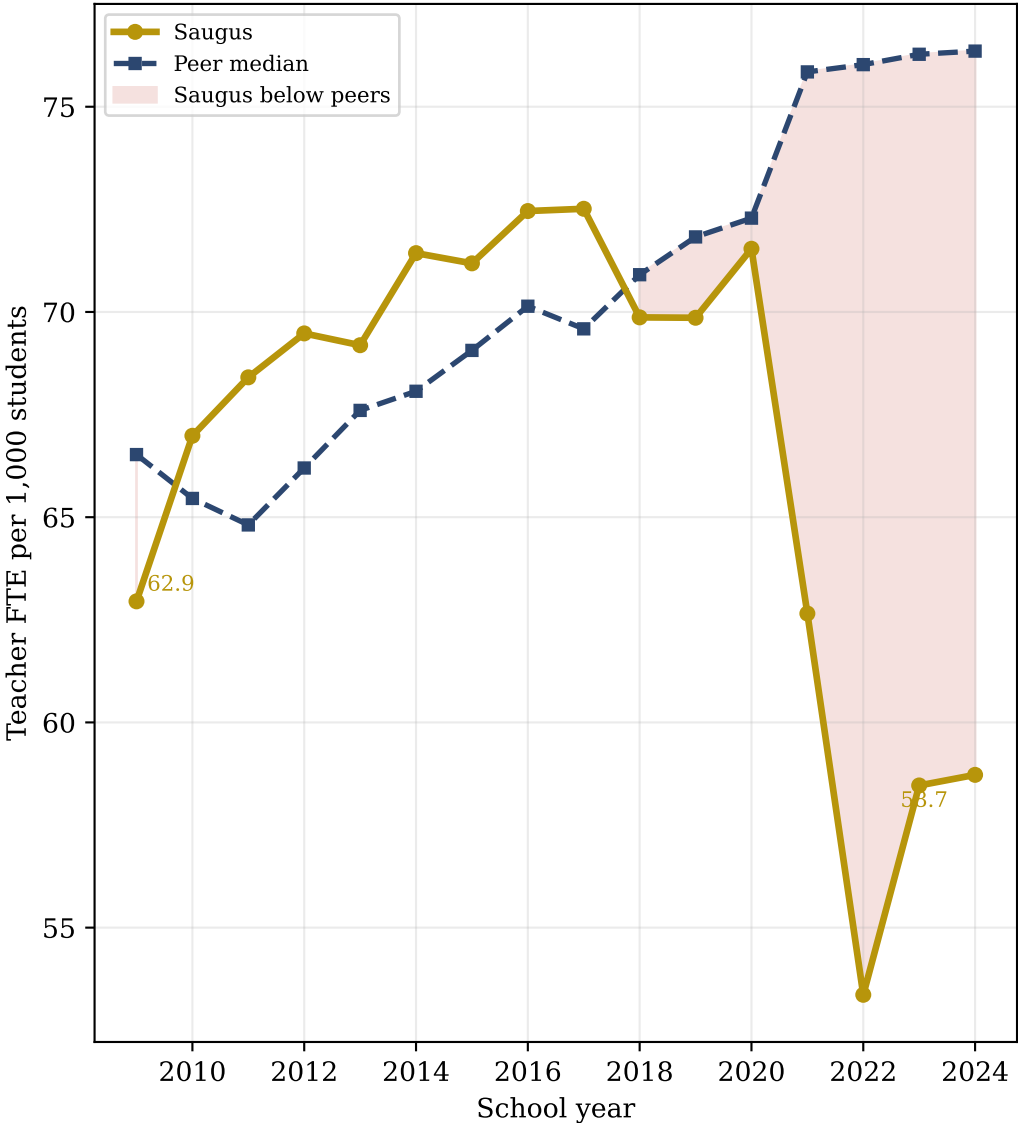
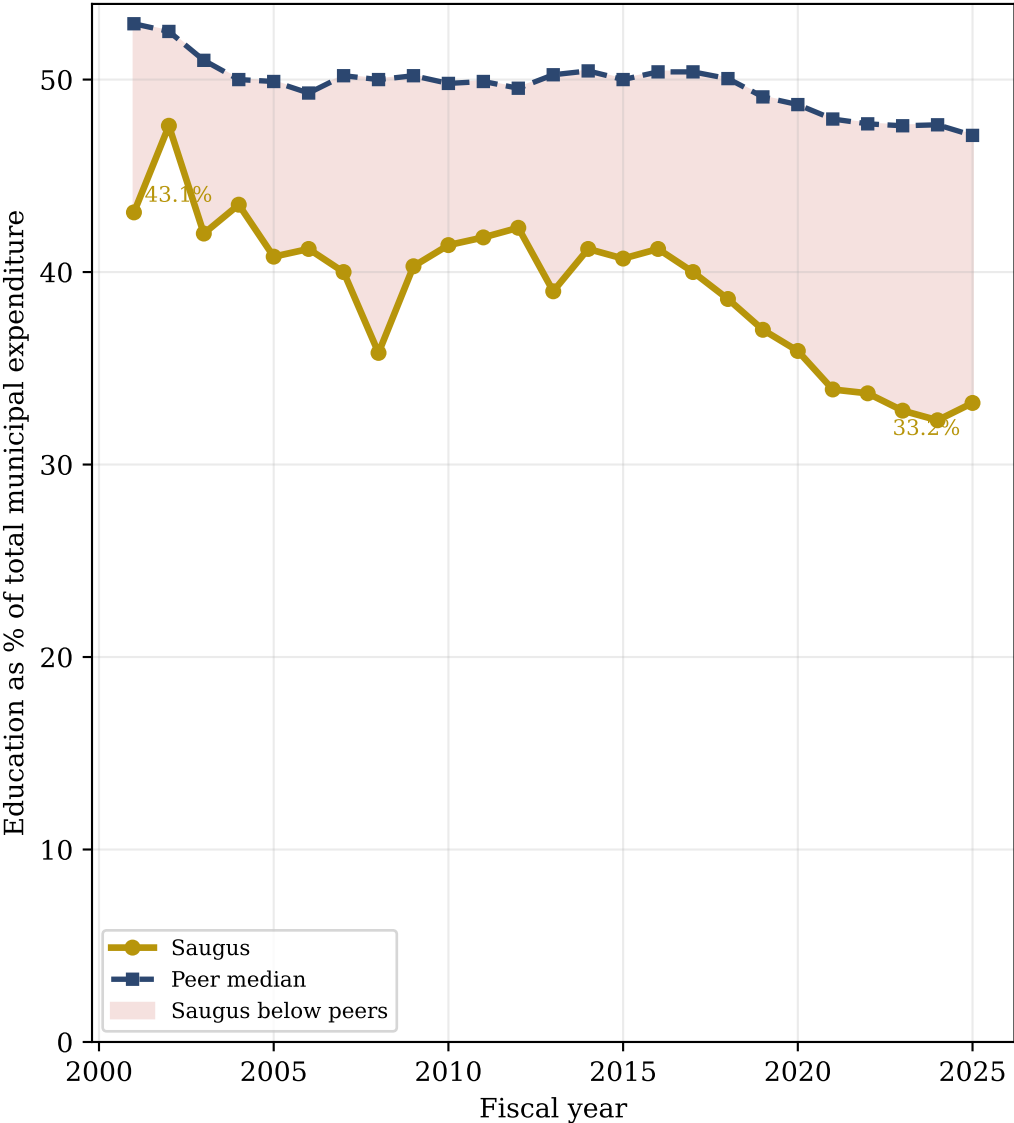
Budget Allocation & Teacher Density — 25-Year Trajectory (FY2001-2025)

Source: MA DLS Schedule A (budget) · MA DESE Staffing (teachers) · Peer comparison: median of 68 demographically similar MA towns (standardized-

Education Budget Share
(Schedule A, general fund)

Mahalanobis $d_M \leq 2\sigma$ on 6 demographic features)

Teacher Density
(DESE Staffing + Enrollment)

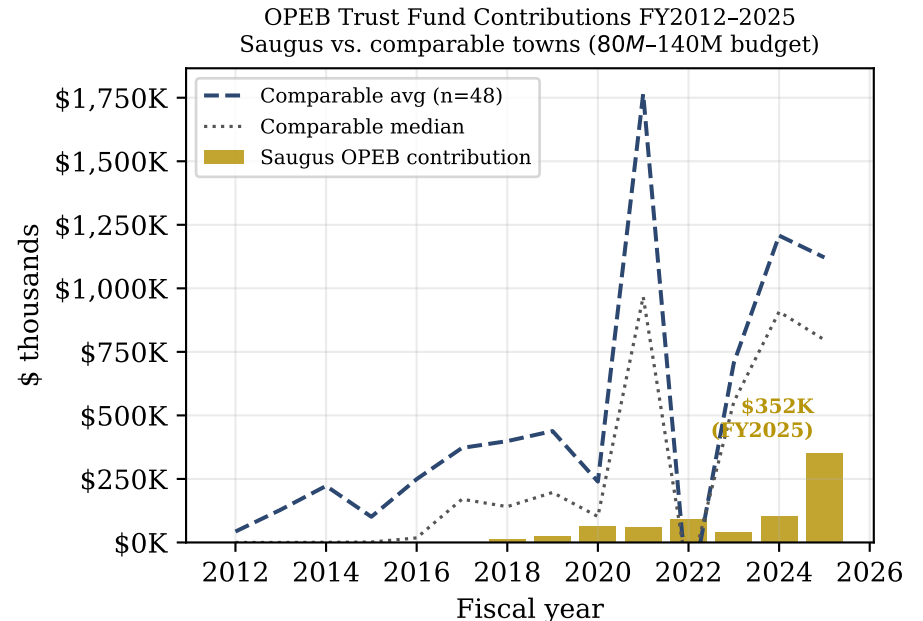
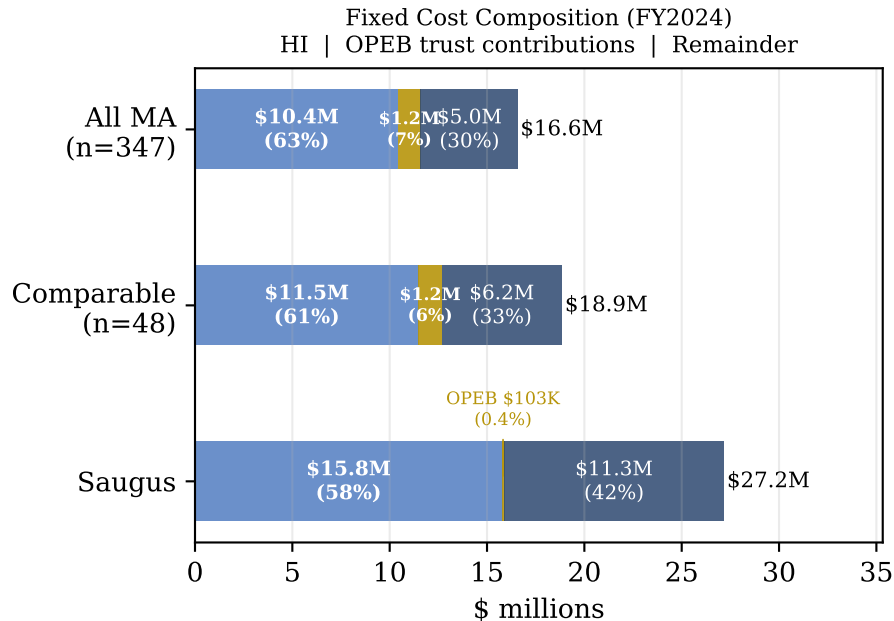
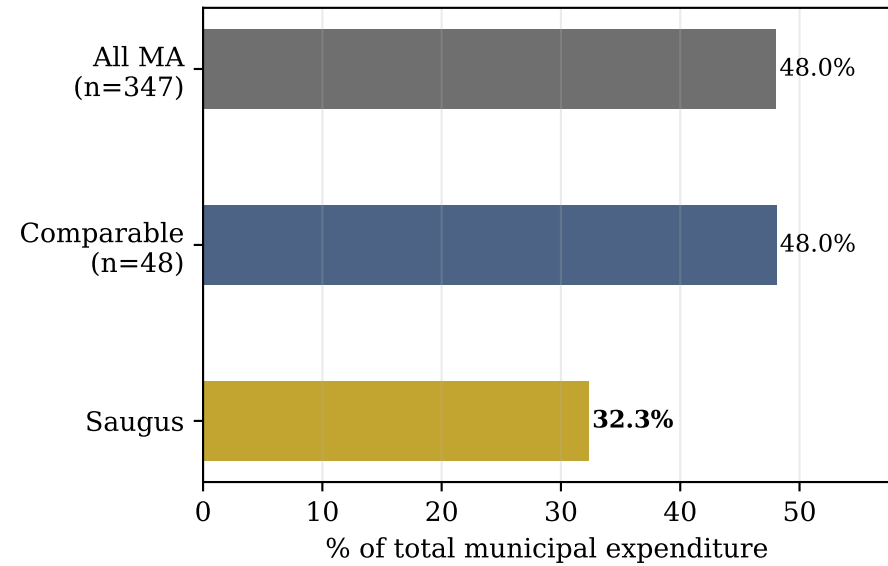
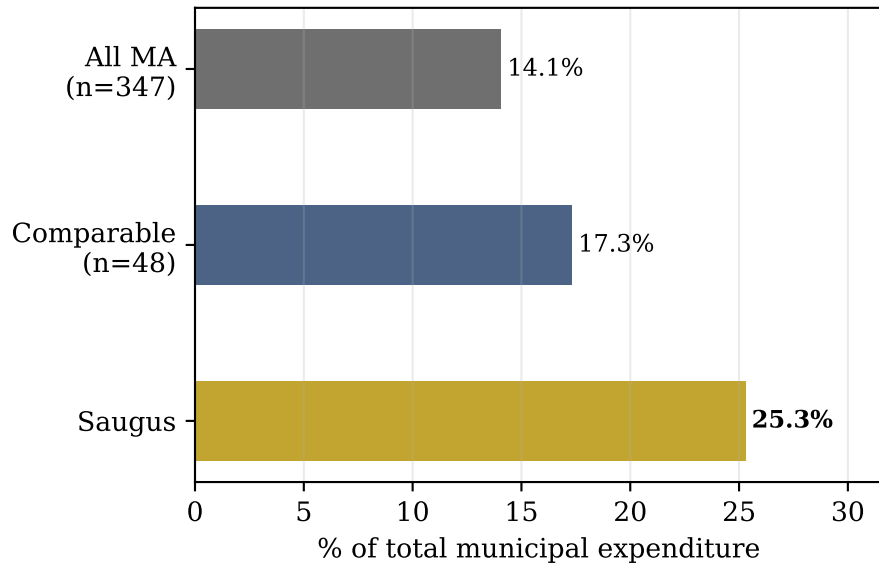


Independent cross-validation (computed in this run): standardized Ridge regression on 219 MA districts (5-fold CV $R^2=0.82$) ranks students chronically absent (10%+) #12 of 19 standardized predictors of MCAS 3-8 ($|\text{standardized } \beta|=0.08$) — a divergence to read with care: a linear model assigns students chronically absent (10%+) a small standardized coefficient while RBP's Exhibit 5 ranks it among the top factors. Some of this is what RBP is built to catch — nonlinear / interaction effects a single linear coefficient misses — but some is mechanical: RBP importance is task-specific and is elevated for any feature on which the subject town is an outlier, and Saugus sits well into the upper tail of the statewide distribution on students chronically absent (10%+). A high RBP rank therefore sharpens Saugus's prediction; on its own it is not evidence that students chronically absent (10%+) is a uniquely powerful driver.

Peer set: median of 68 demographically similar MA towns within standardized-Mahalanobis $d_M \leq 2\sigma$ of Saugus on the 6 demographic features (median_hh_income, equalized_income, low_income_pct, total_enrollment, pct_bachelors_plus, ell_pct). Same similarity rule as the Optimum Profile page; peers derived, not hand-picked; all similar towns, not only better-performing ones.

Fixed Costs: Sources, Scale, and OPEB Prefunding

FY2024 · Comparable = 48 MA towns (80M-140M) · All MA = 347 · Columns sum exactly to total for all 8,119 statewide municipality-years



Health insurance from DLS Schedule A Part 2 & 6 (self-insured municipalities). OPEB & workers comp contributions from Schedule A Part 6 trust fund revenues (municipal trust funds, Revenues). Remainder = pension assessment (Essex Regional Retirement System) + workers comp insurance + unemployment + other. Saugus began OPEB prefunding in FY2016; FY2025 contribution = 352K vs. comparable - townavg 1,121K.

Optimum Profile — What Overachievers Look Like

Each grid shows actionable factor values for Saugus vs over-performer towns. Showing both peer groups tests whether the finding holds across definitions.

Overachievers — 40 MA districts that most consistently beat their demographic prediction

Feature	Saugus	Springfield	North Adams	Lenox	Lawrence	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	32.4%	30.4%	18.3%	29.2%	18.6%	-12.6pp	40
Teachers / 100 students	5.87	7.37	8.67	10.1	7.42	8.08	+2.21	40
Teachers per low-income student	0.126	0.087	0.121	0.414	0.087	0.209	+0.083	40
Spending vs Ch70 minimum (×)	1.03	1.26	1.45	1.54	1.06	1.33	+0.295	40
School spending vs. property wealth	0.068	0.300	0.315	0.063	0.249	0.089	+0.021	40
Teacher share of school \$	0.349	0.291	0.303	0.369	0.313	0.360	+0.011	40
Shown: Springfield, North Adams, Lenox, Lawrence (+ 36 more in median)								

Demographically Similar Overachievers — 14 of those towns within 2σ Mahalanobis distance of Saugus on income, property wealth, poverty, size, education & ELL

Feature	Saugus	Winthrop	Northbridge	Leominster	West Boyls...	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	17.4%	20.5%	22.5%	15.6%	18.4%	-12.8pp	14
Teachers / 100 students	5.87	7.78	7.08	6.89	8.09	8.05	+2.18	14
Teachers per low-income student	0.126	0.209	0.184	0.126	0.257	0.209	+0.083	14
Spending vs Ch70 minimum (×)	1.03	1.15	1.05	1.04	1.30	1.33	+0.296	14
School spending vs. property wealth	0.068	0.089	0.109	0.130	0.100	0.088	+0.021	14
Teacher share of school \$	0.349	0.361	0.390	0.344	0.399	0.364	+0.014	14
Shown: Winthrop, Northbridge, Leominster, West Boylston (+ 10 more in median)								

Similarity metric: standardized Mahalanobis distance $d_M = \sqrt{(\Delta z \cdot R^{-1} \cdot \Delta z^*)}$, z = population z-score and R = correlation matrix of all MA districts on (median_hh_income, equalized_income, low_income_pct, total_enrollment, pct_bachelors_plus_ell_pct). Threshold: $d_M \leq 2$ (2σ). Pre-specified; not tuned to result. Gap = peer median – Saugus. Red = Saugus below peer median, Green = at or above.